

MATERNOVA

Compiled Team Playbook

Master Task Sheets for All Engineering & Management Roles

ROLE: FRONTEND & UI/UX DEVELOPER

Mission Objective: Execute the highly specific, simplified tasks listed below to build your portion of the national award-winning architecture. Check off items as you complete them.

Phase 1: Elderly & Accessibility Overhaul

- ☐ **Big Targets:** Make every clickable button, link, and menu item at least 48x48 pixels so elderly users with shaky hands can easily tap them.
- ☐ **High Contrast:** Remove any light grey text on white backgrounds. Use dark text on light backgrounds (WCAG AAA standards) for poor eyesight.
- ☐ **Screen Reader Labels:** Add hidden ARIA text tags to all images and icons so blind users' phones can read what is on the screen.

Phase 2: ASHA Worker Dashboard

- ☐ **Patient List Layout:** Build a clean, simple list view where an ASHA worker can see all families in their village.
- ☐ **Risk Color Coding:** Add visual tags (Red, Yellow, Green) next to pregnant women's names based on their risk level so workers know who to visit first.
- ☐ **Offline Forms:** Design the health data entry form so it works smoothly even when the phone shows "No Internet."

Phase 3: Voice Interface Elements

- ☐ **Voice Buttons:** Add large, obvious microphone icons next to text inputs to encourage users to speak their symptoms instead of typing.
- ☐ **Language Dropdown:** Create a prominent, easy-to-use dropdown menu for selecting regional languages (connected to Bhashini API).

ROLE: BACKEND DEVELOPER & DATA MANAGER

Mission Objective: Execute the highly specific, simplified tasks listed below to build your portion of the national award-winning architecture. Check off items as you complete them.

Phase 1: API Triage Routing

- ☐ **Infermedica Setup:** Register for the free Infermedica API. Create endpoints to send patient symptoms from the frontend to Infermedica.
- ☐ **Secure Responses:** Format the medical triage results received from Infermedica into clean JSON to send back to the frontend.

Phase 2: Offline Sync & GovTech Integration

- ☐ **Data Queues:** Build a system that saves ASHA worker data locally if the internet is down, and automatically pushes it to the server when the internet reconnects.
- ☐ **DHIS2 Connections:** Map your local database fields to match the standard formats used by the DHIS2 API (the public health software standard).

Phase 3: Emergency Communications (SOS)

- ☐ **Twilio / EmailJS Integration:** Set up a free Twilio account (for SMS) or EmailJS (for emails).
- ☐ **Automated Alerts:** Write backend logic that instantly fires an SMS with GPS coordinates to an emergency contact or ASHA supervisor if a "Fall Detected" signal is received.

ROLE: AI & MACHINE LEARNING ENGINEER

Mission Objective: Execute the highly specific, simplified tasks listed below to build your portion of the national award-winning architecture. Check off items as you complete them.

Phase 1: Visual Intelligence (Google Gemini)

- ☐ **Medical Report OCR:** Program the Gemini API to accept photos of handwritten lab reports and convert them into organized, digital text data.
- ☐ **Pill Recognition:** Write a prompt for the Gemini API that analyzes a photo of a medicine packet to tell the elderly user what the pill is and when to take it.

Phase 2: Vernacular AI (Bhashini API)

- ☐ **Text Translation:** Connect the app to the Indian Government's Bhashini API to instantly translate English medical advice into local languages (Hindi, Tamil, etc.).
- ☐ **Voice-to-Text:** Set up Bhashini's speech recognition so rural users can speak in their dialect and the app converts it to text for processing.

Phase 3: Audio Analysis

- ☐ **Infant Cry Script:** Write a Python script that analyzes the frequency and pitch of an audio file (a baby crying) to classify whether the cry indicates hunger, pain, or sleepiness.

ROLE: MOBILE & SYSTEMS ENGINEER

Mission Objective: Execute the highly specific, simplified tasks listed below to build your portion of the national award-winning architecture. Check off items as you complete them.

Phase 1: Progressive Web App (PWA) Foundation

- ☐ **App Manifest:** Create the `manifest.json` file so Vercel can serve the app as a mobile installable icon on users' home screens.
- ☐ **Service Workers:** Write service worker scripts to cache the website's core files (HTML, CSS, images) so it loads instantly even with zero network coverage.

Phase 2: Hardware Sensor Integration

- ☐ **Accelerometer Listening:** Write JavaScript that runs in the background to monitor the smartphone's accelerometer data (movement speed/angle).
- ☐ **Fall Detection Logic:** Code an algorithm that detects sudden drops or severe impacts, which triggers a countdown for the user to cancel before sending an SOS.

Phase 3: Audio & Camera Pipelines

- ☐ **Camera Access:** Ensure the browser asks for and securely handles camera permissions so elderly users can take photos of pills smoothly.
- ☐ **Microphone Streaming:** Build the audio-capture element that records 5-second clips of infant cries and packages them neatly to send to the AI backend.

ROLE: PROJECT MANAGER & PITCH LEAD

Mission Objective: Execute the highly specific, simplified tasks listed below to build your portion of the national award-winning architecture. Check off items as you complete them.

Phase 1: Timeline & Execution

- ☐ **45-Day Tracker:** Break down the whole project into weekly goals. Check in with all 4 engineers every 3 days to remove blockers.
- ☐ **Scope Control:** Prevent "feature creep." Ensure the team ONLY builds what is on these task sheets and ignores unnecessary distractions.

Phase 2: Field Testing (Crucial for Winning)

- ☐ **Find Beta Testers:** Locate 1-2 actual ASHA workers, nurses, or elderly individuals in your local area to try the app on their phones.
- ☐ **Gather Telemetry:** Write down exactly where they got stuck, what loaded too slowly, and how well the offline mode worked in their hands. Give this feedback to the team.

Phase 3: The Winning Pitch Deck

- ☐ **Structure the Story:** Build the presentation focusing on "The Triad of Vulnerability" (Maternal, Elderly, Infant) and how technology solves their specific problems.
- ☐ **Highlight GovTech:** Make sure the slides explicitly emphasize the integration with ASHA workers, DHIS2, and Bhashini (Juries love this).
- ☐ **Live Demo Prep:** Prepare a flawless, rehearsed live demonstration showing the offline sync and vernacular voice features working in real-time.